



AIMS

African Institute for
Mathematical Sciences
RWANDA

Introduction to Probability and Statistics

Lecture 14: Introduction to the Bayesian Analysis

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- 1 Statistical Bayesian Model
- 2 Decision Theory
- 3 Risks
- 4 Bayes Estimator

Bayesian statistical model

Definition

A **Bayesian statistical model** is a triple $(\mathcal{X}, \mathcal{A}, P_\theta, \pi(\theta), \theta \in \Theta)$ where $\pi(\theta)$ is **prior** distribution of θ .

Observation: $X \sim f(x | \theta)$ (*likelihood*)

➡ $\pi(\theta | x)$ **posterior distribution**.

Continuous version of the **Bayes formula**:

$$\pi(\theta | x) = \frac{f(x, \theta)}{f(x)} = \frac{f(x | \theta)\pi(\theta)}{f(x)}$$

where $f(x)$ is called the **predictive**

$$f(x) = \int_{\Theta} f(x | \theta)\pi(\theta)d\theta$$

From prior to posterior

Example

Let $X_1, \dots, X_n$ be n i.i.d.r.v's from an exponential distribution with parameter $\theta > 0$.

- Likelihood:

From prior to posterior

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$$f(\underline{x} \mid \theta) = (1/\theta)^n \exp \left\{ - \sum_{i=1}^n x_i / \theta \right\}.$$

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$$\pi(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} (1/\theta)^{\alpha+1} \exp\{-\beta/\theta\}, \quad \theta \in \mathbb{R}^+, \quad \alpha, \beta > 0.$$

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- Joint distribution:

$$f(\underline{x}, \theta) = f(\underline{x} \mid \theta) \pi(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} (1/\theta)^{n+\alpha+1} \exp \left\{ -(\beta + \sum_{i=1}^n x_i) / \theta \right\}$$

From prior to posterior

- Predictive:

From prior to posterior

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$$\begin{aligned}f(\underline{x}) &= \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^{+\infty} (1/\theta)^{n+\alpha+1} \exp\{-(\beta + \sum_{i=1}^n x_i)/\theta\} \\&= \frac{\beta^\alpha}{\Gamma(\alpha)} \frac{\Gamma(n+\alpha)}{(\sum_{i=1}^n x_i + \beta)^{n+\alpha}}\end{aligned}$$

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- Posteriori distribution:

From prior to posterior

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- Posteriori distribution:

$$\pi(\theta \mid \underline{x}) = \frac{(\sum_{i=1}^n x_i + \beta)^{n+\alpha}}{\Gamma(n + \alpha)} (1/\theta)^{n+\alpha+1} e^{-(\beta + \sum_{i=1}^n x_i)/\theta}.$$

➡ Inverse gamma distribution with parameter $(n + \alpha, \sum_{i=1}^n x_i + \beta)$.

Decision Model

- $\mathcal{X}$: observation space,
- Θ : state of nature space (parameter space for a statistical problem),
- A : space of **actions** or **decisions** which elements are function of the observation $\delta(x)$.
 - ➡ δ is called a **decision rule** (a statistic or an estimator for a statistical problem),
- $\mathcal{D}$: set of **decisions rules** δ , application from $\mathcal{X}$ to A (all possible estimators),

Loss function

- Let $a = \delta(x)$ be an action.
- Doing inference is choosing a decision rule $\delta \in \mathcal{D}$ about $\theta \in \Theta$, the state of the nature relying on an observation $x \in \mathcal{X}$ given that x and θ are related by $f(x | \theta)$.
- In statistics, the *decision rule* is an *estimator*, the *action* is an *estimate* (estimator value at x).
- To make a *decision*, we need to measure the **loss** incurred when a decision $\delta(x)$ is considered as the *state of the nature* is θ .

Definition

A loss function is a function L from $\Theta \times A$ to $\mathbb{R}$.

➡ $L(\theta, a)$ assess the cost of the decision a when the parameter is θ .

A decision is said to be a “*good decision*”, if the loss associated is null.

Therefore, it is a solution to the equation $L(\theta, \delta(X)) = 0$.

But θ is unknown!

➡ Expectation (average) of the loss: **Frequentist Risk**.

Definition

The **frequentist risk** of a decision rule is the expectation of the loss associated this decision rule:

$$R(\theta, \delta) = E_{\theta}[L(\theta, \delta(X))] = \int_{\mathfrak{X}} L[\theta, \delta(x)] dP_{\theta}(x)$$

Bayes Risk

Definition

The **Bayes risk** is the expectation of the frequentist risk with respect to the prior $\pi(\cdot)$:

$$r(\pi, \delta) = E^{\pi}[R(\theta, \delta)].$$

Remark: $r(\pi, \delta) = \int_{\mathcal{X}} \int_{\Theta} L(\theta, \delta(x)) \pi(\theta | x) f(x) dx d\theta$.

Definition

The expectation of the loss with respect to the posterior distribution is the **Bayesian loss**.

$$\rho(\pi, \delta(x)) = E^{\pi(\cdot|x)}[L(\theta, \delta(x))] = \int_{\Theta} L(\theta, \delta(x)) \pi(\theta | x) d\theta$$

Bayes estimator

Definition

A **Bayes estimator** is a decision rule that minimize the **Bayes risk**:

$$\delta^\pi = \underset{\delta \in \mathcal{D}}{\operatorname{Argmin}} r(\pi, \delta).$$

Since

$$r(\pi, \delta) = E^f[\rho(\pi, \delta(X))].$$



$$(\text{Minimize } r(\pi, \delta)) \iff (\text{Minimize } \rho(\pi, \delta(X))).$$

Bayes estimator

Definition

The function $L(\theta, \delta(x)) = (\theta - \delta(x))^2$ is called the **squared-error loss** or **quadratic loss**.

Proposition

Assuming a squared-error loss, a **Bayes estimator** (bayesian decision) $\delta^\pi(x)$ for θ , associated to a prior π , is the expectation of the posterior distribution:

$$\delta^\pi(x) = E^{\pi(\cdot|x)}(\theta).$$

Proof:

$$\rho(\pi, \delta(x)) = E^{\pi(\cdot|x)}[L(\theta, \delta(x))] = E^{\pi(\cdot|x)}(\theta^2) - 2\delta(x)E^{\pi(\cdot|x)}(\theta) + \delta^2(x).$$

Bayes estimator

Example

- $X_i \mid \theta \sim \mathcal{E}(\theta)$,
- Prior: $\theta \sim \mathcal{IG}(\alpha, \beta)$ (Inverse-gamma distribution),
- Posterior: $\theta \mid \underline{X} \sim \mathcal{IG}(n + \alpha, \sum_{i=1}^n x_i + \beta)$.

Recall:

The expectation of (a r.v.) an **Inverse-gamma distribution** with parameter (a, b) is $b/(a - 1)$.

Bayes estimator

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Recall:

The expectation of (a r.v.) an **Inverse-gamma distribution** with parameter (a, b) is $b/(a - 1).$

Applying the previous proposition, a bayes estimator for θ , assuming a squared-error loss, is therefore:

$$\delta^\pi(x) = (\sum_{i=1}^n x_i + \beta) / (n + \alpha - 1).$$

Bayes estimator

Example

- $X_i \mid \theta \sim \mathcal{B}(N, p)$,
- Prior: $p \sim \mathcal{Beta}(\alpha, \beta)$,

$$\pi(p) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\alpha-1}(1-p)^{\beta-1}, \quad 0 < p < 1, \alpha, \beta > 0.$$

- Posterior: $\theta \mid \underline{X} \sim \mathcal{Beta}(\sum_{i=1}^n X_i + \alpha, nN - \sum_{i=1}^n X_i + \beta)$.

Recall:

The expectation of (a r.v.) a **beta distribution** with parameter (a, b) is $a/a + b$.

Bayes estimator

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Recall:

The expectation of (a r.v.) a **beta distribution** with parameter (a, b) is $a/a + b$.

Applying the previous proposition, a bayes estimator for θ , assuming a squared-error loss, is therefore:

$$\delta^\pi(x) = \frac{\sum_{i=1}^n X_i + \alpha}{nN + \alpha + \beta}.$$